

The aim of numerical analysis is therefore, to provide constructive methods for obtaining answers to such problems in a numerical form.

With the advent of high speed computers and increasing demand for numerical solution to various problems, numerical techniques have become indispensable tools in the hands of engineers and scientists.

The input information is rarely exact since it comes from some measurement or the other and the method also introduces further error. As such, the error in the final result may be due to an error in the initial data or in the method or both. Our effort will be to minimize these errors, so as to get the best possible results. We therefore begin by explaining various kinds of approximations and errors which may occur in a problem and derive some results on error propagation in numerical calculations.

1.2 Accuracy of Numbers

1. *Approximate numbers.* There are two types of numbers: *exact* and *approximate*. *Exact numbers* are 2, 4, 9, 13, $7/2$, 6.45,... etc. But there are numbers such as $4/3$ ($= 1.33333...$), $\sqrt{2}$ ($= 1.414213...$) and π ($= 3.141592...$) which cannot be expressed by a finite number of digits. These may be approximated by numbers 1.3333, 1.4142 and 3.1416, respectively. Such numbers which represent the given numbers to a certain degree of accuracy are called *approximate numbers*.
2. *Significant figures.* The digits used to express a number are called *significant digits (figures)*. Thus each of the numbers 7845, 3.589, and 0.4758 contains four significant figures while the numbers 0.00386, 0.000587, and 0.0000296 contain only three significant figures since zeros only help to fix the position of the decimal point. Similarly the numbers 45000 and 7300.00 have two significant figures only.
3. *Rounding off.* There are numbers with large number of digits, e.g., $22/7 = 3.142857143$. In practice, it is desirable to limit such numbers to a manageable number of digits such as 3.14 or 3.143. This process of dropping unwanted digits is called *rounding off*.
4. *Rule to round off a number to n significant figures:*
 - (i) Discard all digits to the right of the nth digit.
 - (ii) If this discarded number is
 - (a) less than half a unit in the nth place, leave the nth digit unchanged;

- (b) greater than half a unit in the n th place, increase the n th digit by unity;
- (c) exactly half a unit in the n th place, increase the n th digit by unity if it is odd otherwise leave it unchanged.

For instance, the following numbers rounded off to three significant figures are:

7.893 to 7.89	3.567 to 3.57
12.865 to 12.9	84767 to 84800
6.4356 to 6.44	5.8254 to 5.82

Also the numbers 6.284359, 9.864651, and 12.464762 are rounded off to four places of decimal at 6.2844, 9.8646, 12.4648; respectively.

NOTE

Obs. The numbers thus rounded off to n significant figures (or n decimal places) are said to be correct to n significant figures (or n decimal places).

1.3 Errors

In any numerical computation, we come across the following types of errors:

1. *Inherent errors.* Errors which are already present in the statement of a problem before its solution, are called *inherent errors*. Such errors arise either due to the given data being approximate or due to the limitations of mathematical tables, calculators, or the digital computer. Inherent errors can be minimized by taking better data or by using high precision computing aids.
2. *Rounding errors* arise from the process of rounding off the numbers during the computation. Such errors are unavoidable in most of the calculations due to the limitations of the computing aids. Rounding errors can, however, be reduced:
 - (i) by changing the calculation procedure so as to avoid subtraction of nearly equal numbers or division by a small number; or
 - (ii) by retaining at least one more significant figure at each step than that given in the data and rounding off at the last step.

3. *Truncation errors* are caused by using approximate results or on replacing an infinite process by a finite one. If we are using a decimal computer having a fixed word length of four digits, rounding off 13.658 gives 13.66 whereas truncation gives 13.65.

For example, if $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \infty = X$ (say)

is replaced by $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} = X'$ (say), then the truncation error is $X - X'$.

Truncation error is a type of algorithm error.

4. *Absolute, Relative, and Percentage errors.* If X is the true value of a quantity and X' is its approximate value, then $|X - X'|$ i.e., $|\text{Error}|$ is called the *absolute error* E_a .

The *relative error* is defined by $E_r = \left| \frac{X - X'}{X} \right|$ i.e., $\left| \frac{\text{Error}}{\text{True value}} \right|$

and the percentage error is $E_p = 100E_r = 100 \left| \frac{X - X'}{X} \right|$.

If $\bar{X}$ be such a number that $|X - X'| \leq \bar{X}$, then $\bar{X}$ is an upper limit on the magnitude of absolute error and measures the *absolute accuracy*.

NOTE

Obs. 1. The relative and percentage errors are independent of the units used while absolute error is expressed in terms of these units.

Obs. 2. If a number is correct to n decimal places then the error $= \frac{1}{2}10^{-n}$. For example, if the number is 3.1416 correct to 4 decimal places, then the error $= \frac{1}{2}10^{-4} = 0.00005$.

1.4 Useful Rules for Estimating Errors

To estimate the errors which creep in when the numbers in a calculation are truncated or rounded off to a certain number of digits, the following rules are useful.

If the approximate value of a number X having n decimal digits is X' , then

1. Absolute error due to truncation to k digits

$$= |X - X'| < 10^{n-k}$$

2. Absolute error due to rounding off to k digits

$$= |X - X'| < \frac{1}{2} 10^{n-k}$$

3. Relative error due to truncation to k digits

$$= \left| \frac{X - X'}{X} \right| < 10^{1-k}$$

4. Relative error due to rounding off to k digits

$$= \left| \frac{X - X'}{X} \right| < \frac{1}{2} 10^{1-k}$$

NOTE

Obs. 1. If a number is correct to n significant digits, then the maximum relative error $\leq \frac{1}{2} 10^{-n}$. If a number is correct to d decimal places, then the absolute error $\leq \frac{1}{2} 10^{-d}$.

Obs. 2. If the first significant figure of a number is k and the number is correct to n significant figures, then the relative error $< 1/(k \times 10^{n-1})$.

Let us verify this result by finding the relative error in the number 864.32 correct to five significant figures.

Here $k = 8$, $n = 5$ and

Absolute error $\geq 0.01 \times \frac{1}{2} = 0.005$.

$\therefore$ Relative error

$$\begin{aligned} &\leq \frac{0.005}{864.32} = \frac{5}{864320} = \frac{1}{2 \times 86432} < \frac{1}{2 \times 80000} \\ &= \frac{1}{2 \times 8 \times 10^4} < \frac{1}{8 \times 10^4} \text{ i.e., } \frac{1}{k \times 10^{n-1}}. \end{aligned}$$

Hence the result is verified.

EXAMPLE 1.1

Round off the numbers 865250 and 37.46235 to four significant figures and compute E_a , E_r , E_p in each case.

Solution:

(i) Number rounded off to four significant figures = 865200

$$E_a = |X - X_1| = |865250 - 865200| = 50$$

$$E_r = \left| \frac{X - X_1}{X} \right| = \frac{0.00235}{37.46235} = 6.27 \times 10^{-5}$$

$$E_p = E_r \times 100 = 6.27 \times 10^{-3}$$

(ii) Number rounded off to four significant figures = 37.46

$$\therefore E_a = |X - X_1| = |37.46235 - 37.46000| = 0.00235$$

$$E_r = \left| \frac{X - X_1}{X} \right| = \frac{0.00235}{37.46235} = 6.27 \times 10^{-5}$$

$$E_p = E_r \times 100 = 6.27 \times 10^{-3}$$

EXAMPLE 1.2

Find the absolute error if the number $X = 0.00545828$ is

- (i) truncated to three decimal digits.
- (ii) rounded off to three decimal digits.

Solution: We have $X = 0.00545828 = 0.545828 \times 10^{-2}$

- (i) After truncating to three decimal places, its approximate value

$$X' = 0.545 \times 10^{-2}$$

$$\begin{aligned} \therefore \text{Absolute error} &= |X - X'| = 0.000828 \times 10^{-2} \\ &= 0.828 \times 10^{-5} < 10^{-2-3} \end{aligned}$$

This proves rule (1).

- (ii) After rounding off to three decimal places, its approximate value

$$X' = 0.546 \times 10^{-2}$$

$$\begin{aligned} \therefore \text{Absolute error} &= |X - X'| \\ &= |0.545828 - 0.546| \times 10^{-2} \\ &= 0.000172 \times 10^{-2} = 0.172 \times 10^{-5} \end{aligned}$$

which is $< 0.5 \times 10^{-2-3}$. This proves rule (2).

EXAMPLE 1.3

Find the relative error if the number $X = 0.004997$ is

- (i) truncated to three decimal digits
- (ii) rounded off to three decimal digits.

Solution: We have $X = 0.004997 = 0.4997 \times 10^{-2}$

- (i) After truncating to three decimal places, its approximate value $X = 0.499 \times 10^{-2}$.

$$\begin{aligned} \therefore \text{Relative error} &= \left| \frac{X - X'}{X} \right| = \left| \frac{0.4997 \times 10^{-2} - 0.499 \times 10^{-2}}{0.4997 \times 10^{-2}} \right| \\ &= 0.140 \times 10^{-2} < 10^{-3} \end{aligned}$$

This proves rule (3).

- (ii) After rounding off to three decimal places, the approximate value of the given number

$$X' = 0.500 \times 10^{-2}$$

$$\begin{aligned} \therefore \text{Relative error} &= \left| \frac{X - X'}{X} \right| = \left| \frac{0.4997 \times 10^{-2} - 0.500 \times 10^{-2}}{0.4997 \times 10^{-2}} \right| \\ &= 0.600 \times 10^{-3} = 0.06 \times 10^{-3+1} \end{aligned}$$

which is less than $0.5 \times 10^{-3+1}$. This proves rule (4).

Exercises 1.1

1. Round off the following numbers correct to four significant figures: 3.26425, 35.46735, 4985561, 0.70035, 0.00032217, and 18.265101.
2. Round off the number 75462 to four significant digits and then calculate the absolute error and percentage error.
3. If 0.333 is the approximate value of $1/3$, find the absolute and relative errors.
4. Find the percentage error if 625.483 is approximated to three significant figures.
5. Find the relative error in taking $n = 3.141593$ as $22/7$.
6. The height of an observation tower was estimated to be 47 m, whereas its actual height was 45 m. Calculate the percentage relative error in the measurement.